(i) $f(x) \geq 0,\ \forall x$ (ii) $\int_{-\infty}^{\infty} f(x)dx = 1$ is called the probability density function (p.d.f) of X.

Remark:

(i) Every integrable function satisfying the above two conditions is a probability density function of some random variable X

(ii) The probability that the value of X lies between two points 'a' and 'b' is $P(a < X < b) = \int_a^b f(x)dx$

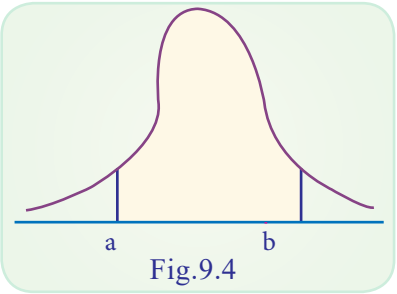


Fig.9.4

(iii) If X is discrete random variable then for any real x, $P(X = x)$ need not be zero. However in the case of continuous random variable $P(X = x) = 0$ holds always. $P(X - a) = \int_a^a f(x)dx = 0$

(iv) If X is a continuous random variable then for any $a < b$ $P(a < X < b) = p(a \leq X < b) = p(a < X \leq b) = p(a \leq X \leq b)$

Example 9.3

A continuous random variable X has probability density function given by $f(x) = \begin{cases} Ax^3, & 0 < x < 1 \\ 0, & otherwise \end{cases}$. Find A.

Solution:

Since $f(x)$ is a p.d.f

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^1 Ax^3 dx = 1$$

$$\Rightarrow \quad \frac{A}{4} = 1$$

$$\Rightarrow \quad A = 4$$

NOTE

Total probability of continuous random variable is 1 within the certain interval.

Example 9.4

Verify whether the following function is a probability density function

$$f(x) = \begin{cases} \frac{2x}{9}, & 0 < x < 3 \\ 0, & elsewhere \end{cases}$$

Solution :

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^3 \frac{2x}{9} dx$$